

researchers to see patterns in data, find trends, clean and classify data, bridge gaps between datasets, and make highly accurate predictions from complex data. However, the potential long-term impacts of AI on research culture and the reliance on digital technologies for the research process and environment are still not fully understood.

While AI has the potential to enhance research productivity, there is a need for caution in its use and further research is required to fully understand its implications.

AI is most commonly used to assist with specific, narrow problems in research, such as identifying patterns and increasing the speed and scale of data analysis, as well as generating new hypotheses. Many respondents from a study titled ‘Speeding up to keep up: exploring the use of AI in the research process’ viewed the rise of web searching and online journal repositories as a positive development, making it easier to keep up with the fast-moving research landscape. Some participants noted that AI could streamline literature searching by sifting through large numbers of abstracts and selecting those that are relevant. Tools such as Paper Digest were reported to be used by some research teams to do the “heavy lifting” of research. However, participants also acknowledged that these tasks could be time-consuming and tedious. One participant noted that such skills are often integral to certain professions, such as accounting and radiology, and suggested that replacing these skills with AI requires a deeper understanding of the skills involved. The goal, therefore, should not be to simply “deskill” these professions, but rather to better understand and replace the skills that are being automated.

Earlier, it was mentioned that AI could help researchers break free from narrow, individual research tasks. This, in turn, could allow them to deliver impact activities more efficiently. According to one participant, academic research typically takes too long to reach beyond academia, and AI’s ability to build connections could accelerate the process. Another participant suggested that AI could enable the “vision of the open-source community applied to AI.” However, some participants cautioned against the potential negative and invasive uses of these infrastructures once established. There was also concern that over-reliance on AI could render academics generalists, which could be both positive and negative. While it could help break down barriers between disciplines, some participants expressed concerns about working across fields they did not train in. Despite this, most participants viewed AI positively for increasing the potential for interdisciplinary collaboration and engagement with the public. One participant noted that